The area of a circle

$S = \pi r^2$ — where the formula comes from, and why the radius is squared.

LESSON 97–99 | 3 × 40 MIN | MATEMATIKA 6, §21 | S7 15

LESSON CLOCK

25' | 30' | 35' | 25' | 5

Where the formula comes from	25 min
Using it	30 min
Backwards	35 min
Doubling the radius	25 min
Homework	5 min

LEARNING OBJECTIVES

- Use $S = \pi r^2$ to find the area of a disc.
- Find the radius from the area.
- Explain why doubling the radius quadruples the area.
- Solve practical problems about circular regions.

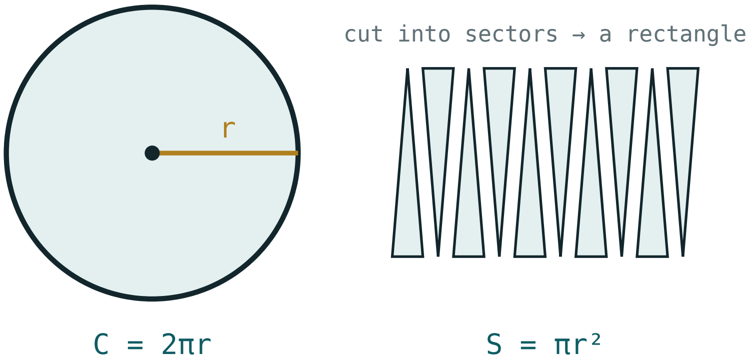
TEXTBOOK

National	pp. 281–290
Cambridge	Stage 7, pp. 148–154
Workbook	Exercise 15.2

01 Explanation

Where the formula comes from

Cut a disc into many thin sectors and lay them alternately point-up and point-down: they form a shape close to a rectangle.



A disc cut into sectors and rearranged

SIDE OF THE RECTANGLE	LENGTH
the long side	half the circumference, πr
the short side	the radius, r
so the area	$\pi r \cdot r = \pi r^2$

$$S = \pi r^2$$

THE THINNER THE SECTORS, THE BETTER THE RECTANGLE

With eight sectors the shape is lumpy; with sixty it is almost exactly a rectangle. The formula is what that process arrives at.

Using it

R	R^2	AREA ($\Pi = \frac{22}{7}$)	AREA ($\Pi = 3.14$)
7	49	154	153.86
14	196	616	615.44
10	100	314.29	314
21	441	1386	1384.74

SQUARE THE RADIUS, NOT ΠR

πr^2 means $\pi \cdot r \cdot r$, not $(\pi r)^2$. With $r = 7$ the area is 154, not 484.

Backwards

GIVEN AREA	$R^2 = S \div \Pi$	R
154	*49*	*7*
616	*196*	*14*
314	*100*	*10*
1386	*441*	*21*

DIVIDE BY Π, THEN TAKE THE SQUARE ROOT

Two steps, in that order. Stopping after the division gives r^2, which is a common half-done answer.

Doubling the radius

R	CIRCUMFERENCE	AREA
7	*44*	*154*
14	*88*	*616*
doubled	doubled	four times

LENGTHS SCALE BY K, AREAS BY K^2

A pizza of twice the diameter is four times as much pizza. It is the same rule met with rectangles, and it applies to every shape.

02 Worked examples

EXAMPLE 1 **Find the area of a disc of radius $7\ cm$, taking $\pi = \frac{22}{7}$.**

1. $S = \pi r^2$
2. $= \frac{22}{7} \cdot 49$
3. $= 154\ cm^2$

   Square units.

ANSWER

$154\ cm^2$

EXAMPLE 2 **A disc has area $616\ cm^2$. Find its radius.**

1. $r^2 = S \div \pi = 616 \div \frac{22}{7}$
2. $= 196$
3. $r = \sqrt{196} = 14$ cm.

ANSWER

$14\ cm$

EXAMPLE 3 **A circular pond of radius *14 m* is to be covered with netting. Find the area needed.**

1. $S = \frac{22}{7} \cdot 196$
2. $= 616$ m².
3. Netting is sold by the square metre, so 616 m² is the order.

ANSWER

$616\ m^2$

03 Interactive model

Cut a paper disc into twelve sectors and rearrange them on the board; the near-rectangle makes πr^2 obvious in a way no derivation can.

Area, forwards and backwards

The area of a disc is:

- $2\pi r$
- πr^2
- πd
- πr

With $r = 7$ and $\pi = \frac{22}{7}$:

- 44
- 154
- 49
- 484

With $r = 10$ and $\pi = 3.14$:

- 31.4
- 62.8
- 314
- 100

An area of 154 gives r^2 equal to:

- 7
- 49
- 154
- 22

So the radius is:

- 7
- 49
- 14
- 22

Doubling the radius multiplies the area by:

- 2
- 3
- 4
- 8

And the circumference by:

- 2
- 4
- 8
- 1

An answer in cm for an area is:

- fine
- wrong units
- a radius
- a diameter

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Area	Yuza	Площадь
Disc	Doira	Круг
Radius	Radius	Радиус
Square units	Kvadrat birlik	Квадратные единицы
To square	Kvadratga ko‘tarish	Возвести в квадрат
Square root	Kvadrat ildiz	Квадратный корень
Scale factor	O‘xshashlik koeffitsiyenti	Коэффициент подобия
Circular region	Doiraviy soha	Круговая область

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$S = \pi r^2$ means:

- $(\pi r)^2$
- $\pi \cdot r \cdot r$
- $2\pi r$
- $\pi + r^2$

A disc of radius 14 has area:

- 88
- 616
- 196
- 308

To find r from S you:

- divide by π
- divide by π then take the root
- take the root
- multiply

A disc of area 314 has radius:

- 10
- 100
- 50
- 31.4

Trebling the radius multiplies the area by:

- 3
- 6
- 9
- 27

Area is measured in:

- cm
- cm^2
- cm^3
- degrees

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $r = 7$: the area ($\pi = \frac{22}{7}$)

   ANSWER $154\ cm^2$

2. $r = 14$: the area

   ANSWER $616\ cm^2$

3. $r = 10$: the area ($\pi = 3.14$)

   ANSWER $314\ cm^2$

4. $r = 21$: the area ($\pi = \frac{22}{7}$)

   ANSWER $1386\ cm^2$

5. $r = 5$: the area ($\pi = 3.14$)

   ANSWER $78.5\ cm^2$

6. The formula for the area of a disc

   ANSWER πr^2

7. The units of an area

   ANSWER cm^2

Medium · the standard question

7 PROBLEMS

1. Area 154: the radius

   ANSWER 7

2. Area 616: the radius

   ANSWER 14

3. Area 314: the radius

   ANSWER 10

4. A pond of radius 14 m: the netting needed

   ANSWER 616 m²

5. $d = 14$: the area

   ANSWER $154\ cm^2$

6. $d = 28$: the area

   ANSWER $616\ cm^2$

7. Doubling r multiplies the area by

   ANSWER 4

Hard · stretch and reason

7 PROBLEMS

1. A ring between radii 7 and 14

   ANSWER 462 cm²

2. A ring between radii 10 and 6 ($\pi = 3.14$)

   ANSWER 200.96 cm²

3. A disc of circumference 44: its area

   ANSWER 154 cm²

4. A disc of area 154: its circumference

   ANSWER 44 cm

5. Trebling the radius multiplies the area by

   ANSWER 9

6. A pizza of 30 cm across against two of 15 cm: which is more?

   ANSWER The one of 30 cm, by twice

7. Why is the radius squared?

   ANSWER Area has two directions, and both scale with r

07 Homework

Homework — 5 tasks

Square the radius before multiplying by π, and write cm^2.

1. Find the area of discs of radius $3.5\ cm$ and $21\ cm$.
2. Find the area of a disc of diameter $28\ cm$.
3. A disc has area $1386\ cm^2$. Find its radius.
4. Find the area of a ring between circles of radii 14 and 7 cm.
5. Explain in one sentence why doubling the radius quadruples the area.